Artificial and Computational Intelligence
AIMLCZG557

BITS Pilani

Pilani|Dubai|Goa|Hyderabad

ACI Team

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BITS Pilani

Pilani|Dubai|Goa|Hyderabad

AIMLCZG557 – CS#4
Designing Heuristics & Introduction to Local Search Algorithms

Agenda for CS #4

1) Recap of CS #3
2) Designing Heuristics
3) Heuristics from Relaxed problems, Pattern DB
4) Use case: Reference Paper
5) Local Search Algorithms
6) Q&A!

Introducing Heuristics: The "Good Enough" Guiding Star

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What is a Heuristic?

A practical rule of thumb or educated guess that estimates the cost from current state to the goal. Derived from Greek "heuriskein" – "to find" or "to discover."

Key Insight

Prioritize exploration of paths that **seem most promising** rather than exploring all possibilities blindly.

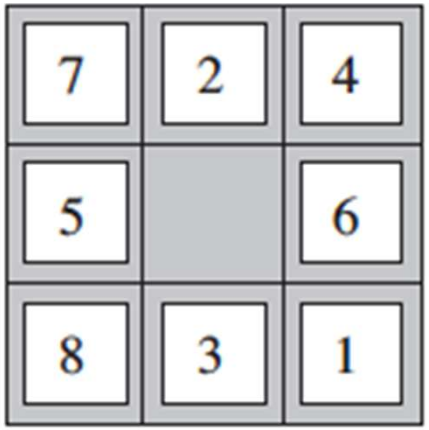
Like using a compass and map in a maze instead of wandering aimlessly

Heuristics Design

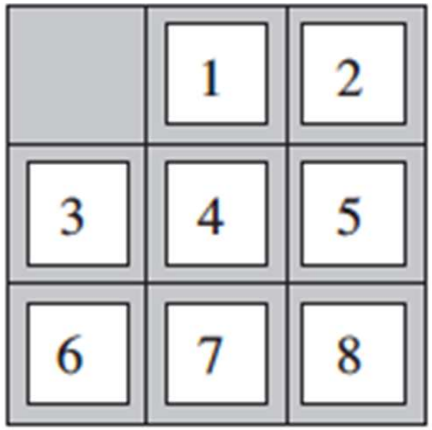
- As heuristic functions determine the search strategies and thereby their complexities,
  - it is important that we identify better heuristic functions
  - While maintaining Admissible and Consistency constraints
- A good heuristic design can be selected using:
  - Effective Branching Factor
  - Generating admissible heuristics from relaxed problems
  - Generating admissible heuristics from sub problems
  - Learning Heuristics from Experience

Activity: Heuristics

Provide atleast 2 heuristics for this problem!



Start State



Goal State

h_1 (n) = number of misplaced tiles, admissible

h_2 (n) = sum of Manhattan distances of tiles from their respective goal positions, admissible

Test:

h_1 (S) = ? 8

h_2 (n) = ? 3+1+2+2+2+3+3+2 = 18

$$| x_1 - x_2 | + | y_1 - y_2 |$$

Need of Heuristics

- Humans rarely search blindly.
- We use educated guesses.
- AI systems also need estimates to guide search.
- These estimates are called **heuristics**.

Example: While driving, we choose roads that appear to move us closer to the destination.

Real-Life Examples:

- Google Maps suggests shortest routes 🗺️
- Chess engines evaluate board positions ♟️
- Doctors infer diseases from symptoms 🩺
- Robots estimate distance to targets 🤖

The AI Challenge: Navigating Infinite Possibilities

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Core Task

AI must solve complex problems by searching through vast, often infinite state spaces. For example, the 15-puzzle has 1.3 trillion possible states to explore.

Uninformed Search

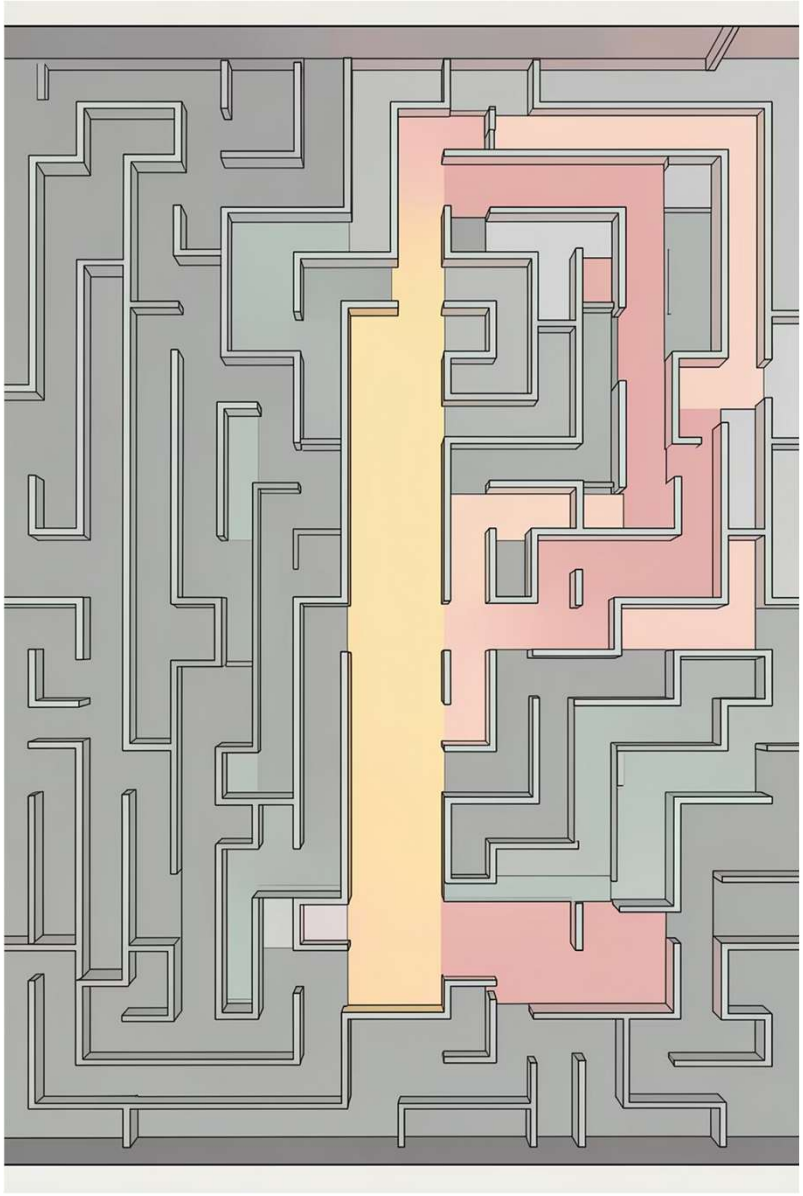
BFS and DFS explore systematically but inefficiently, like navigating a maze blindfolded and exploring millions of unnecessary paths.

Computational Barrier

Exponential growth makes brute-force search impossible for real-world AI problems.

We require "informed" search strategies that use domain knowledge to guide the AI efficiently toward solutions.

The Power of Informed Search: A* and Beyond



01

Best-First Search

Expands nodes based on heuristic evaluation function

02

Cost Function

f(n) = g(n) + h(n) where g(n) is actual cost from start and h(n) is estimated cost to goal

03

A* Algorithm

Guarantees optimal path if heuristic is admissible (never overestimates true cost)

Pruning the Search Tree: Efficiency Gains

1

Intelligent Pruning

Good heuristics eliminate irrelevant branches from exploration

2

Performance Boost

Significantly reduces nodes explored, saving time and memory

3

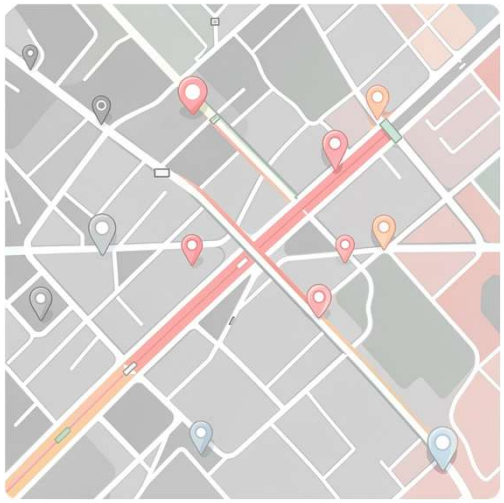
Trade-off Spectrum

Perfect heuristic: O(d).

Zero heuristic: O(b^d)

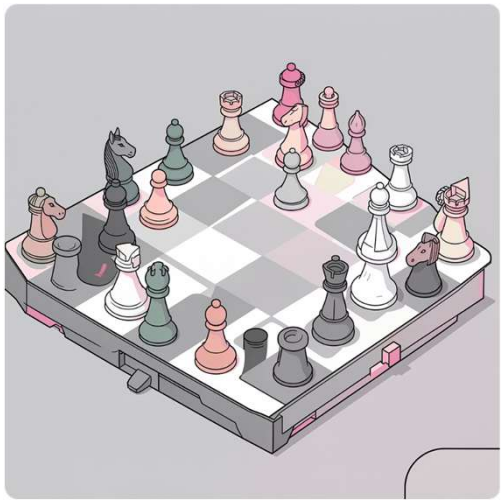
"The better the heuristic, the less time" – Ellen Walker, Hiram College

Real-World AI Applications: Where Heuristics Shine



Pathfinding

GPS navigation systems use Manhattan or Euclidean distance heuristics for route optimization



Game AI

Chess engines like Deep Blue evaluate board positions using sophisticated heuristic functions

Robotics

Robot navigation and motion planning rely on heuristics to find efficient collision-free paths

Logistics

Optimizing delivery routes and resource allocation in complex scheduling problems

Advanced Heuristic Techniques: Pushing the Boundaries

IDA* (Iterative Deepening A*)

Memory-bounded version of A* for large state spaces, combining depth-first search with cost limits

SMA* (Simplified Memory-bounded A*)

Further memory optimization by dropping nodes when memory is full, retaining best paths

Combining Heuristics

Merging multiple heuristic functions through maximum operation for improved accuracy

Learning Heuristics

Deriving heuristics from relaxed problems, subproblems, pattern databases, or past experience

Mastering Heuristic Design

Foundation

Understanding heuristic properties: admissibility, consistency, and dominance relationships

Design Skills

Creating effective heuristic functions for novel AI problems and domains

Implementation

Building and analyzing search algorithms like A*, IDA*, and SMA* variants

Your Role: *Become architects of intelligent search, building more efficient and capable AI systems that can tackle real-world complexity.*

Heuristic Design

- Effective Branching Factor (b∗)
  - If the algorithm generates N number of nodes, and the solution is found at depth d, then

$$N + 1 = 1 + (b^*) + (b^*)^2 + (b^*)^3 + \ldots + (b^*)^d$$

- Good Heuristics
  - Accurate, Fast to compute, Admissible, Consistent, Informative
- Notion of Relaxed Problems
  - A relaxed problem removes some constraints from the original problem, making it easier to solve. The cost of the relaxed problem becomes a lower bound on the original problem cost.
- Generating Admissible Heuristics

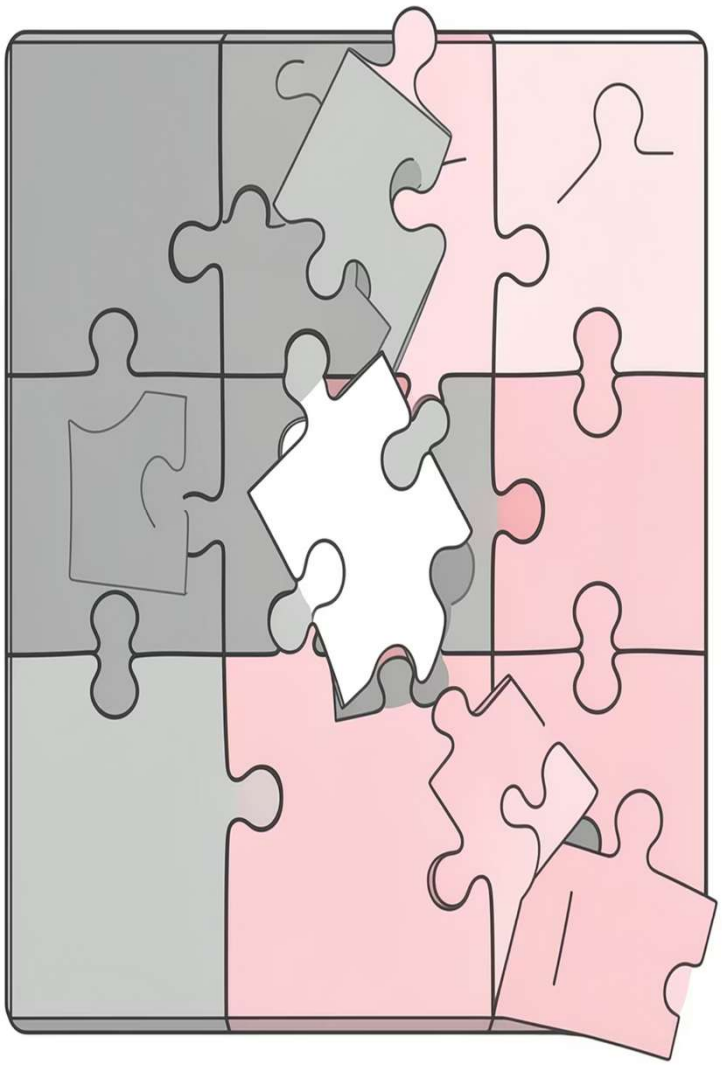
How do we design these heuristics?

- Relaxed Problems: We create a simpler version of the game by removing rules.
  - For example, if tiles could teleport or move through each other, the puzzle would be easy to solve. The cost of solving this "easy" version becomes our heuristic for the "hard" version.
- Pattern Databases: We solve a small part of the puzzle first (like just the first four tiles) and save every possible configuration and its cost in a database. When the AI is searching, it just looks up the answer in the table.
- Landmarks: Imagine you are using a GPS. It's too hard to calculate the path between every single street corner in the world. Instead, the system picks "landmark" cities and pre-calculates the distances to them.
  - By comparing where you are to a landmark, the AI can estimate the distance to your destination.

Generating Heuristics from Relaxed Problems

Idea: Solve a simplified version of the original problem where some constraints are removed. The cost to solve the relaxed problem becomes an admissible heuristic for the original problem.

1. **Original Problem**
   Move tiles to form specific configuration with movement constraints
2. **Relaxed Problem 1**
   Ignore tile movement constraints—tiles can move anywhere.
   Heuristic: Number of misplaced tiles
3. **Relaxed Problem 2**
   Allow tiles to move through each other. Heuristic:
   Manhattan distance (sum of distances from goal position)



Key Benefit: Provides a lower bound on true cost, ensuring optimality in A* search while dramatically reducing explored nodes.

Generating Heuristics with Landmarks

Idea

Identify a set of "landmark" states that are easy to reach from many other states. The heuristic h(n) is based on the distance to the nearest landmark, providing informed estimates for large, complex graphs.

Example: Road Networks

Landmarks could be major cities or highway interchanges. The heuristic for a current location n is the shortest known distance from *n* to any landmark.

Advantage

Can provide more informed estimates than simple relaxed problems, especially in large graphs where traditional heuristics struggle.

Learning from Experience

Can an AI learn to be a better searcher on its own? Yes!

- Metalevel Learning: This is when the AI “thinks about its own thinking”. It looks at the search paths it took and learns to avoid "dead ends" or unhelpful steps in the future.
- Feature Learning: An AI can solve thousands of random puzzles and look for patterns. It might notice that certain features, like how many tiles are out of order, are great predictors of the final cost and combine them into a single mathematical formula to guide its search.

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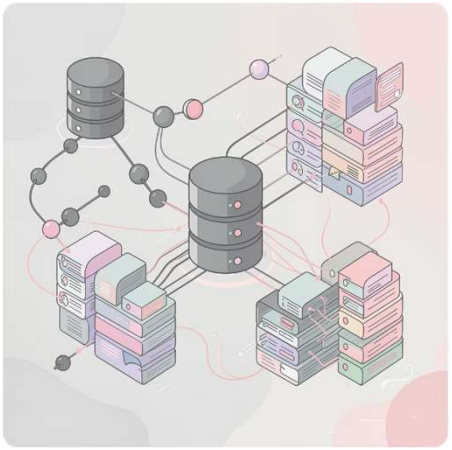
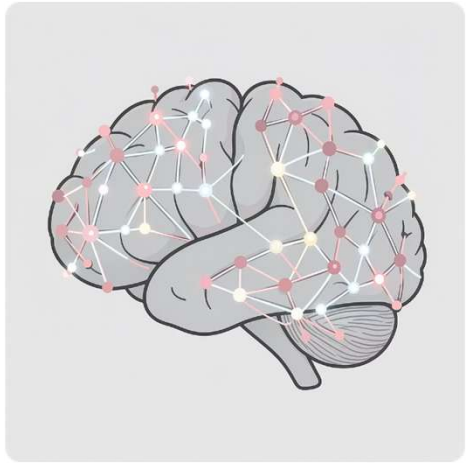
Learning Heuristics from Experience

Instead of manually designing heuristics, machine learning discovers patterns from data, learning to predict costs from solved problem instances.



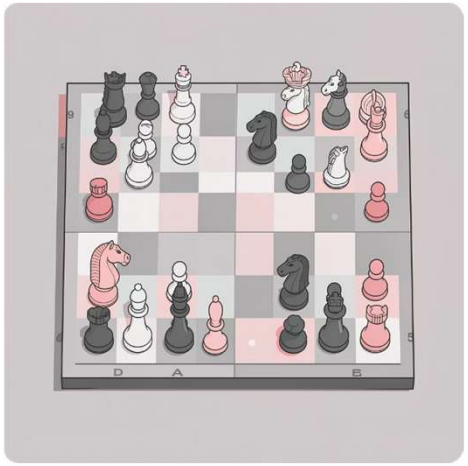
Pattern Databases

Pre-compute optimal solutions for smaller subproblems and use them to estimate costs for larger problems efficiently.



Neural Networks

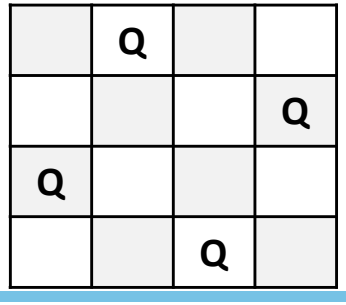
Train deep neural networks to predict the cost to the goal based on the current state representation.



Game Applications

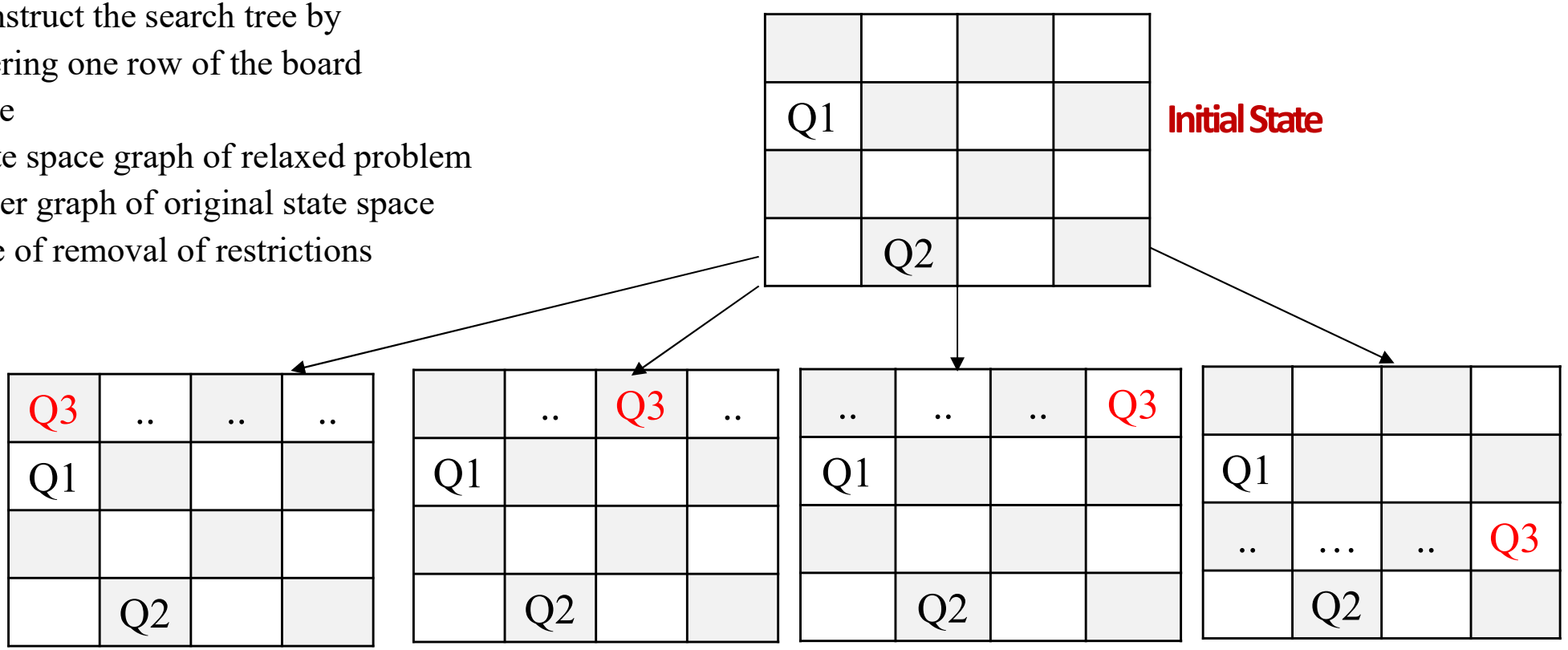
Learning heuristics for complex games like Go or Chess, where manual design is nearly impossible.

N-Queen



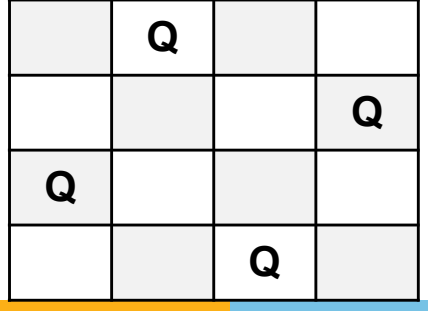
Goal State

- Construct the search tree by considering one row of the board at a time
- State space graph of relaxed problem is a super graph of original state space because of removal of restrictions

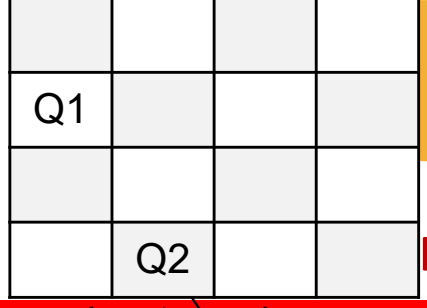


Initial State	Possible Actions	Transition Model	Goal Test	Path Cost	No.Of.States
< Xi , Yi >	Place in any non-occupied row in board		isValid Non-Attacking	Transition + Valid Queens	n!

N-Queen



Goal State



Initial State

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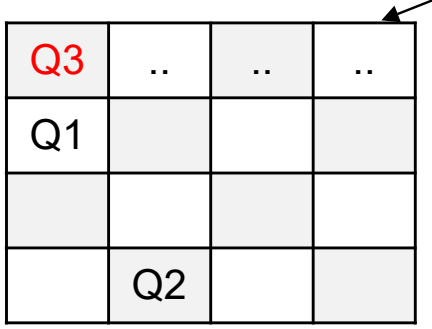
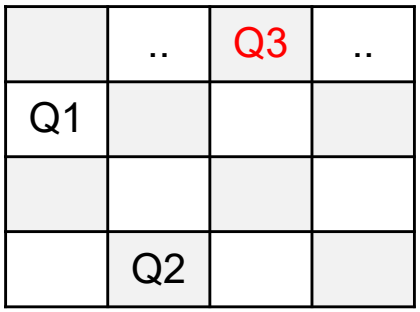
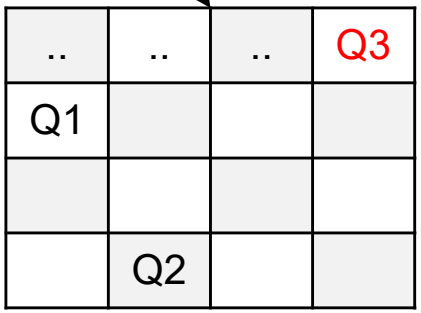
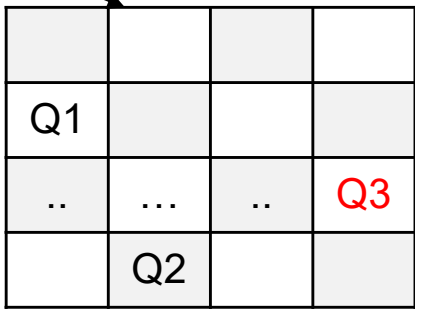
(Q1, Q2)
(Q1, Q3)
(Q2, Q3)







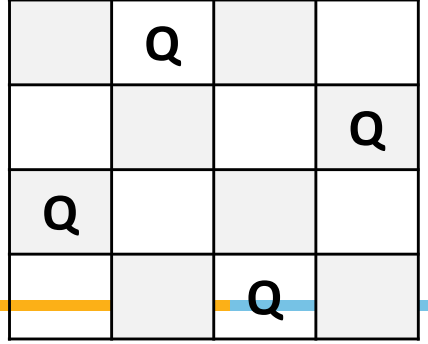


Possible Heuristic Design:

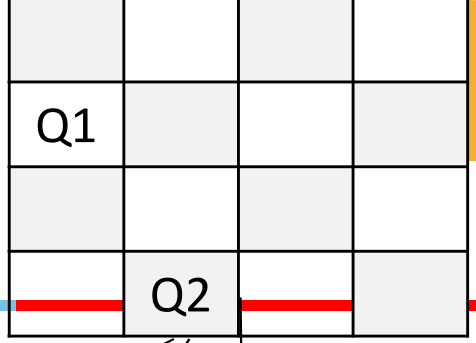
H1(n) : Number of Non conflicting Pairs of Queen (MAX)	2	3	3	3
H2(n) : Number of Conflicting Pairs of Queen (MIN)	1	0	0	0
H3(n) : Number of safest Queen (MAX)	1	3	3	3

BITS Pilani, Deemed to be University under Section 3 of UGC Act, 1956

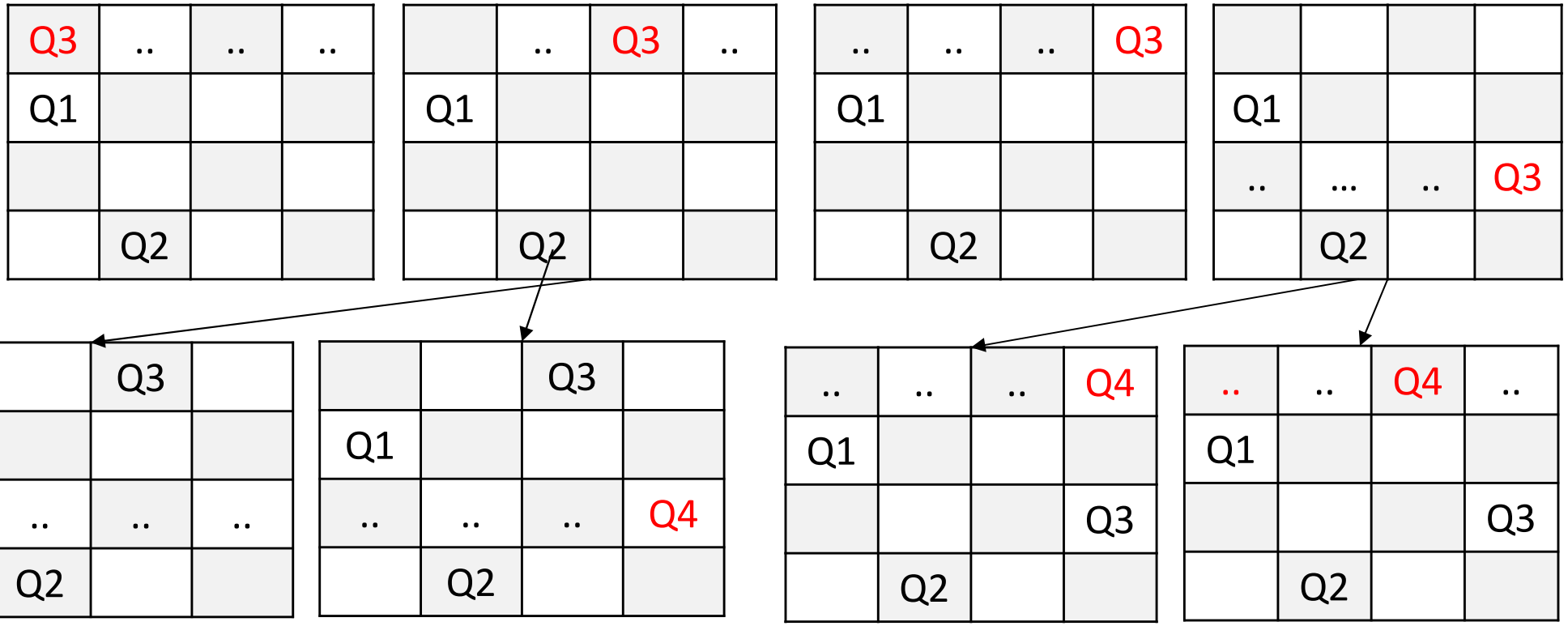
N-Queen



Goal State



Initial State



N-Tile

Initial State	Possible Actions	Transition Model	Goal Test	Path Cost	No.Of.States
<LOC, ID>	Move Empty to near by Tile		ID=LOC+1	Transition + Positional + Distance+ Other approaches	9!

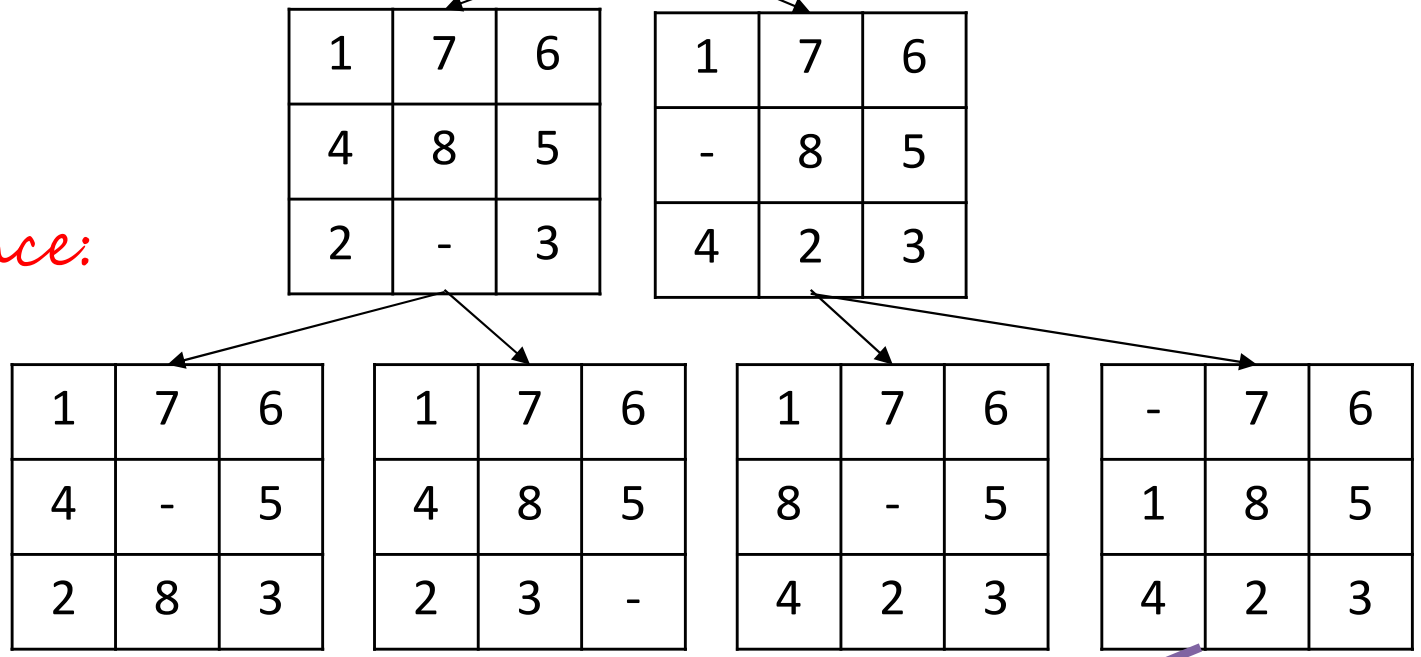
N-Tile

	1	2	3
1	-	1	2
2	3	4	5
3	6	7	8

Goal State

	1	7	6
	4	8	5
	-	2	3

Initial State



Manhattan Distance:

$|x1-x2| + |y1-y2|$

Possible Heuristic Design:

H1(n) : Manhattan distance of Empty tile	0
H2(n) : Manhattan distance of all labelled tile	1:2 2:3 3:3 4:2 5: 0 6:4 7:2 8:2 18
H3(n) : Number of Misplaced tile (MIN)	7

Heuristic Design: Use case

Reference paper

- The paper uses A* and Greedy Best-First Search (GBFS) as the two main search algorithms to evaluate *learned heuristics* on several grid-based domains, specifically:
  - Sokoban
  - Maze with Teleports
  - Sliding Tile Puzzle
- These domains naturally represent states as grids, making them suitable for convolutional neural networks.
- The authors train neural networks to produce heuristic values and then use those heuristics inside A* and GBFS to solve the problems.

Grid Domains Used in the Paper

- **Sokoban:**
  - A warehouse puzzle where an agent pushes boxes to target positions.
- **Maze with Teleports:**
  - A maze where the agent must reach a goal while optionally using teleport pairs.
- **Sliding Tile Puzzle:**
  - A classic N-puzzle where tiles are moved into a target arrangement.
  - All three can be encoded as 2D grids and processed by convolutional neural networks.

Neural Network as the Heuristic Function

The paper represents the heuristic as:

$$h\,(s, \theta)$$

where:

- s = current grid state,
- θ = learned neural network parameters,
- $h\,(s, \theta)$ = predicted heuristic value.

For grid domains, the network contains:

- 7 convolution layers,
- 4 CoAt (Convolution + Attention) blocks,
- Mean pooling,
- A fully connected layer producing one scalar heuristic value.

How A* Uses the Learned Heuristic?

A* evaluates each node using:

$$f(n) = g(n) + h(n)$$

where:

- $g(n)$:actual path cost from the start,
- $h(n)$:*neural network estimate*,
- $f(n)$:total estimated solution cost.

At each step:

1. Generate successor states.
2. Compute $h\ (s, \theta)$ for each successor.
3. Compute $f(n)$.
4. Expand the node with the smallest $f(n)$.

How GBFS Uses the Learned Heuristic?

GBFS ignores the accumulated cost and chooses the state with the smallest heuristic:

$$f(n) = h(n)$$

At each step:

- Generate successors.
- Predict heuristic values.
- Expand the state with the lowest predicted value.

✓ GBFS is often faster but does not guarantee optimal solutions.

Partially Observable

Irrelevant Paths

Goal Oriented -> Expected Goal

Memory limitation

Local Search & Optimization

Greedy choice

Next Best Successor

Optimization is central to Artificial Intelligence. Many AI problems require finding the best solution among a vast number of possibilities!

Path vs State Optimization

- Previous topics: path to goal is solution to problem
  - systematic exploration of search space.
- This topic (local search): a *state* is solution to the problem!
  - for some problems path is irrelevant.
  - E.g., 8-queens
- In many optimization problems, the path to the goal is irrelevant; the goal state itself is the solution
- State space = set of "complete" configurations
  - Find configuration satisfying constraints, e.g., n-queens
- In such cases, we can use local search algorithms
  - Keep a single "current" state, try to improve it

Local search and optimization

- Local search
  - Keep track of single current state
  - Move only to neighboring states
  - Ignore paths
- Advantages:
  - Use very little memory
  - Can often find reasonable solutions in large or infinite (continuous) state spaces.
- "Pure optimization" problems
  - All states have an objective function
  - Goal is to find state with max (or min) objective value
  - Does not quite fit into path-cost/goal-state formulation
  - Local search can do quite well on these problems.

Optimization Models

- Local search algorithms – Search in the state-space in the neighborhood of current position until an optimal solution is found
  - Hill Climbing Algorithm and its variants
  - Simulated Annealing
  - Local Beam Search
- Population based algorithms – Instead of single instance in state-space, use multiple parallel instances (i.e., population) in finding the optimal solution
  - Genetic Algorithms, Ant colony optimization

Local Search

- Local search algorithms focus on improving a single current state rather than constructing complete search trees.
- They are especially useful when only the final solution matters, not the path to the solution.
- Applications:
  - Hyperparameter tuning in machine learning
  - Feature selection
  - Robotics path planning
  - Scheduling and timetabling
  - VLSI design

Optimization Models

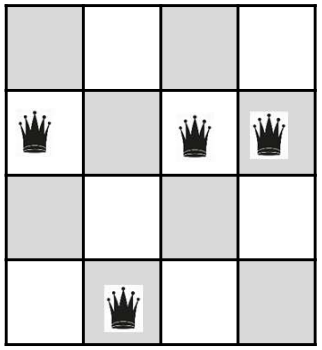
- **Goal** : Navigate through a state space for a given problem such that an optimal solution can be found
- **Objective** : Minimize or Maximize the objective evaluation function value
- **Scope** : Local
- **Objective Function** : Fitness Value evaluates the goodness of current solution
- **Local Search** : Search in the state-space in the neighbourhood of current position until an optimal solution is found

Single Instance Based	Multiple Instance Based
Hill Climbing ✅	Genetic Algorithm ✅
Local Beam Search ✅	Ant Colony Optimization ✅
Simulated Annealing	Particle Swarm Optimization
Tabu Search	

Local Search Terminology

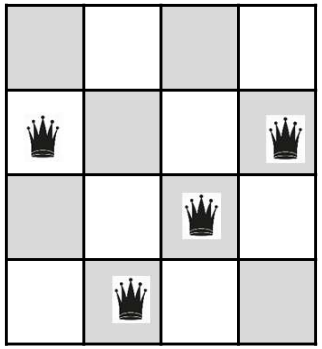
- **Local Search:** Search in the state-space in the neighborhood of current position until an optimal solution is found.

Feasible State/Solution

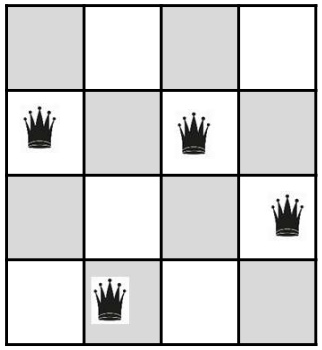


V1	V2	V3	V4
2	4	2	2

Neighbour States

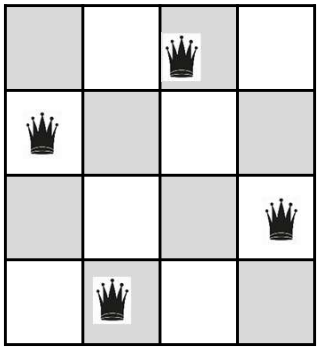


V1	V2	V3	V4
2	4	3	2



V1	V2	V3	V4
2	4	2	3

Optimal Solution

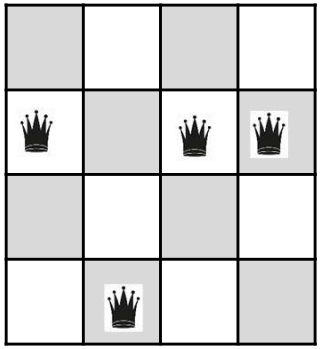


V1	V2	V3	V4
2	4	1	3

Fitness Value: h(n)= Number of conflicting pairs of queens

h(n)= 4 h(n)= 4 h(n)= 2 h(n)= 0

Local Search Expansion



V1	V2	V3	V4
2	4	2	2

1NN:

(2-1) (2-3) (2-4)
(4-1) (4-2) (4-3)
(2-1) (2-3) (2-4)
(2-1) (2-3) (2-4)

2NN:

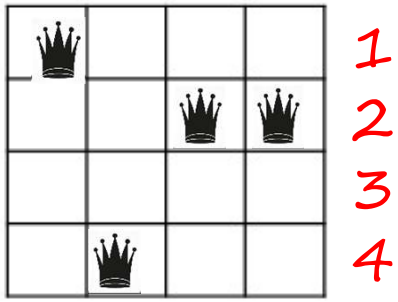
V1 V2 → 9
V1 V3 → 9
V1 V4 → 9
V2 V3 → 9
V2 V4 → 9
V3 V4 → 9

Hill Climbing

1. Select a random state
2. Evaluate the fitness scores for all the successors of the state

h(n) = Number of non-conflicting pairs of queens in the board.

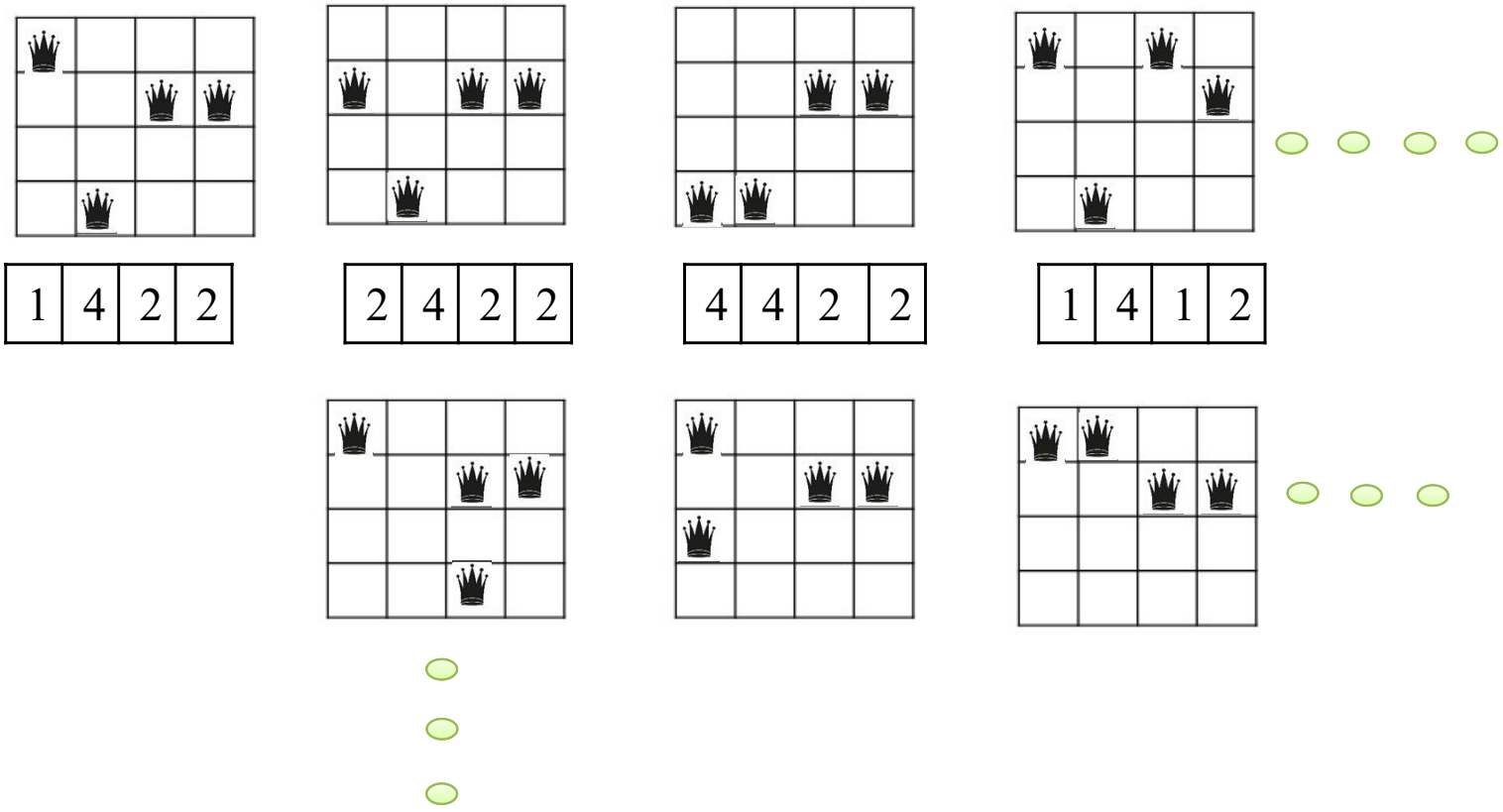
N Q1-Q2
N Q1-Q3
N Q1-Q4
N Q2-Q3
Y Q2-Q4
Y Q3-Q4



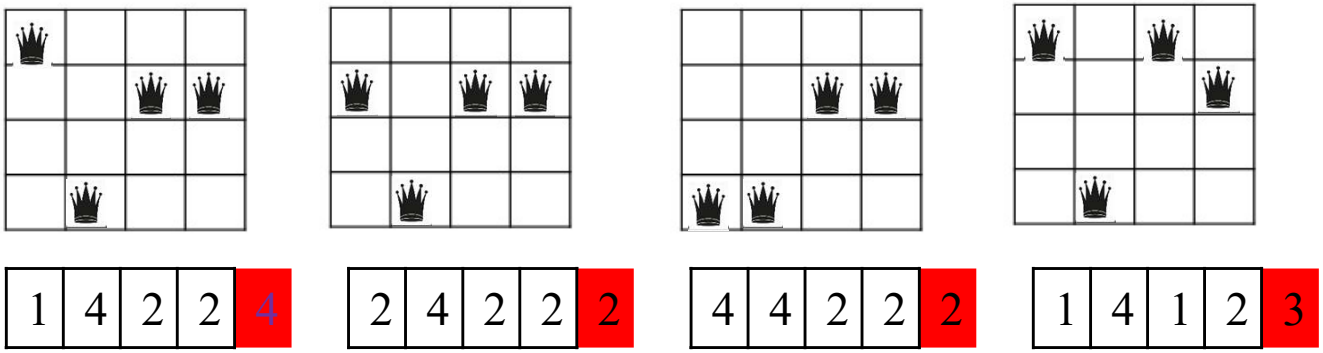
1	4	2	2	4

Hill Climbing

1. Select a random state
2. Evaluate the fitness scores for all the successors of the state



Hill Climbing



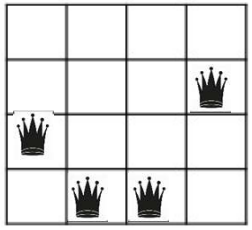
Local Maxima → Random Restart

Hill Climbing

```
function HILL-CLIMBING(problem) returns a state that is a local maximum
    current ← problem.INITIAL
    while true do
        neighbor ← a highest-valued successor state of current
        if VALUE(neighbor) ≤ VALUE(current) then return current
        current ← neighbor
```

Hill Climbing with Random Restart

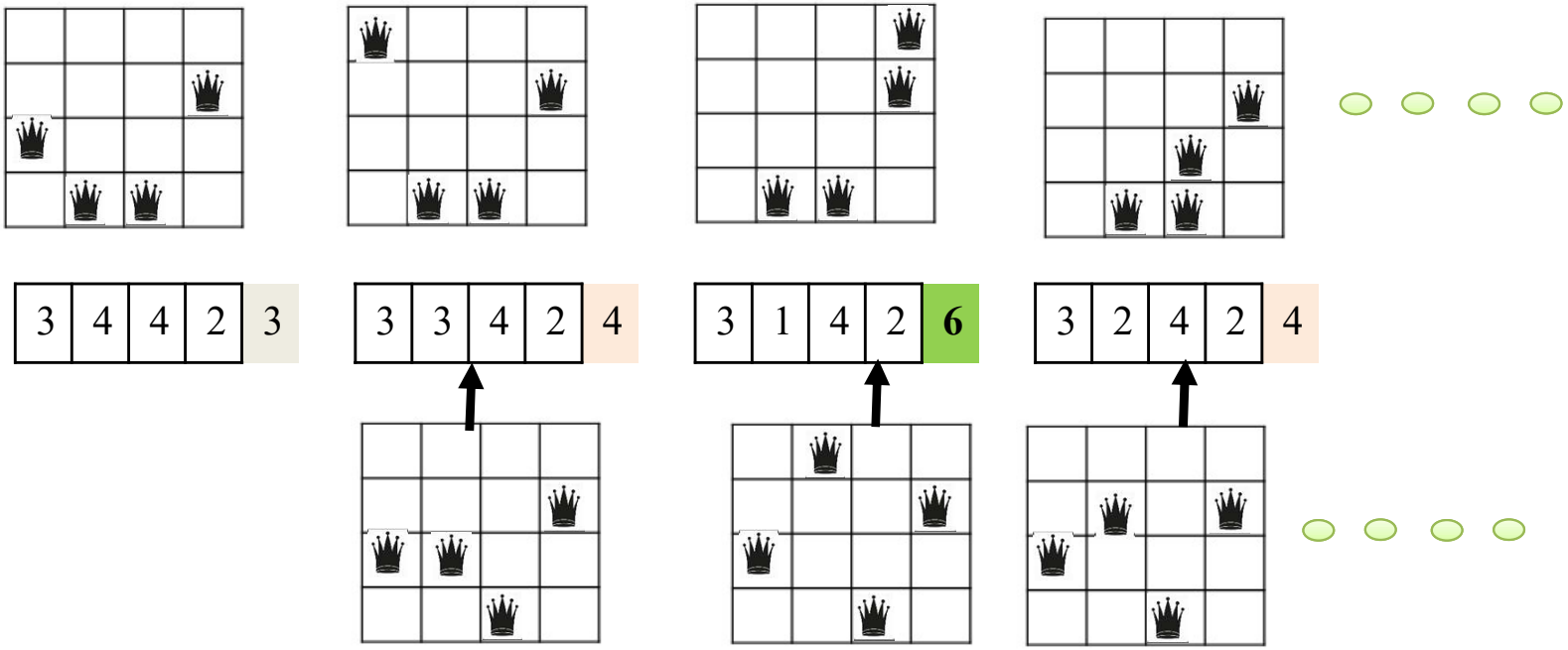
1. Select a random state
2. Evaluate the fitness scores for all the successors of the state
3. Calculate the probability of selecting a successor based on fitness score
4. Select the next state based on the highest probability
5. Repeat from Step 2



3	4	4	2	3

Hill Climbing with Random Restart

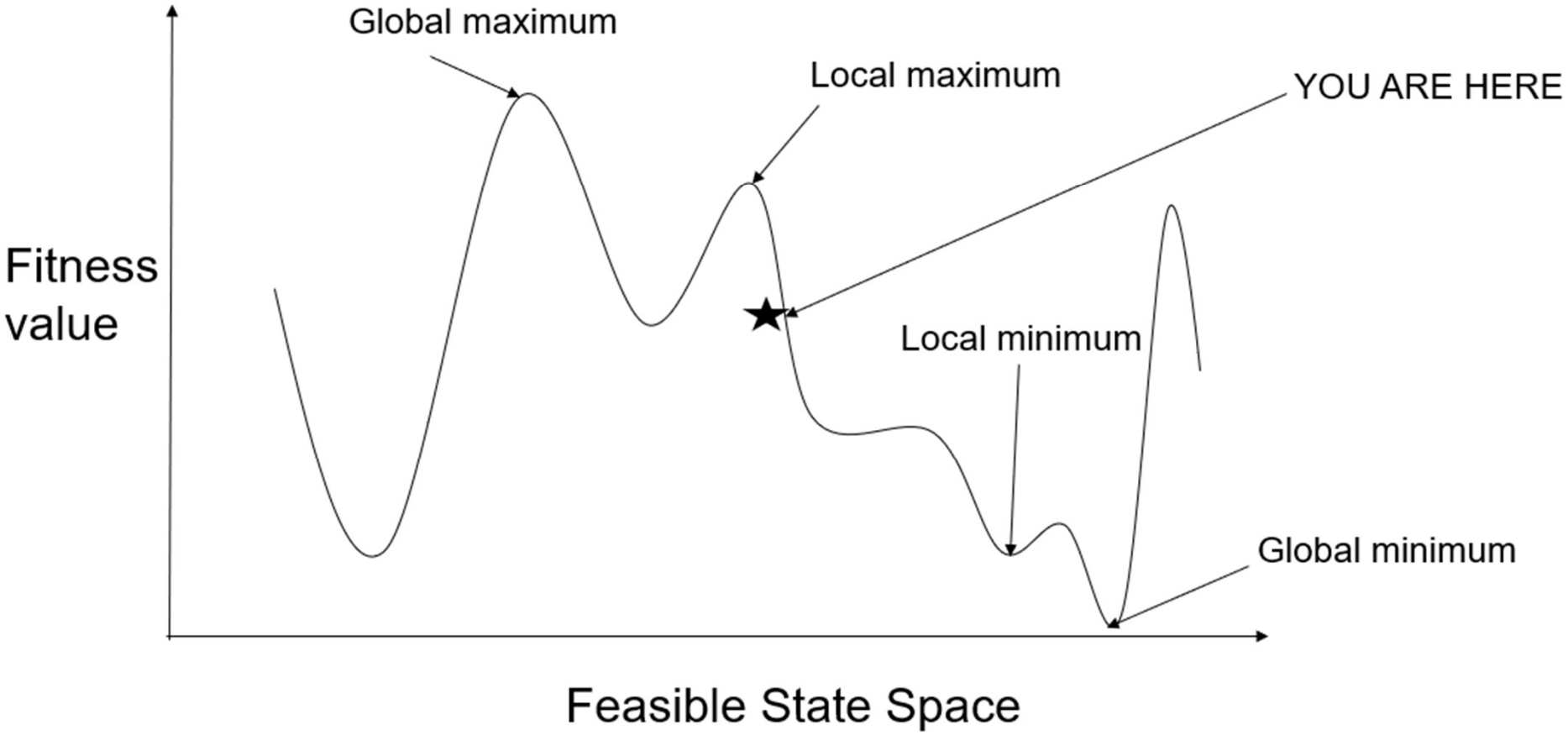
1. Select a random state
2. Evaluate the fitness scores for all the successors of the state
3. Calculate the probability of selecting a successor based on fitness score
4. Select the next state based on the highest probability
5. Repeat from Step 2



Hill Climbing with Random Restart (When to Stop?)

- Termination Condition 1: When no successor has a better fitness value
- Termination Condition 2: When we attained Global Maxima or Global minima
- Termination Condition 3: When a predefined value of ‘k’ denoting the number of restart is reached.

Hill Climbing

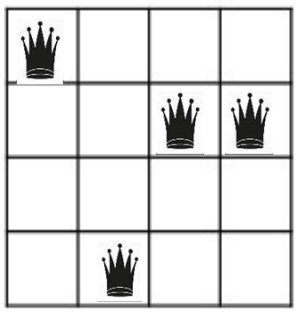


Stochastic Hill Climbing

$next \leftarrow$ a randomly selected successor of $current$
$\Delta E \leftarrow next.\text{VALUE} - current.\text{VALUE}$
if $\Delta E > 0$ **then** $current \leftarrow next$
else $current \leftarrow next$ only with probability $e^{\Delta E/T}$

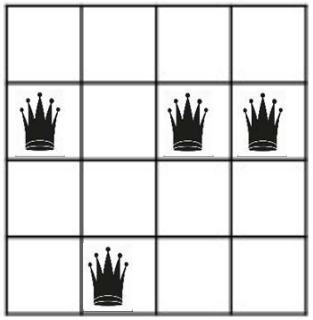
Stochastic Hill Climbing

1. Select a random state
2. Evaluate the fitness scores for all the successors of the state
3. Calculate the *probability* of selecting a successor based on fitness score
4. Select the next state based on the highest probability
5. Repeat from Step 2



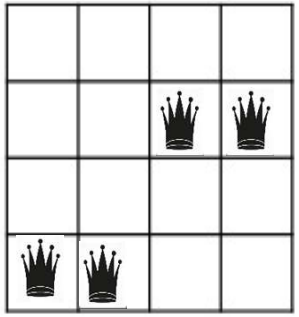
1	4	2	2	4

Probability=0.08



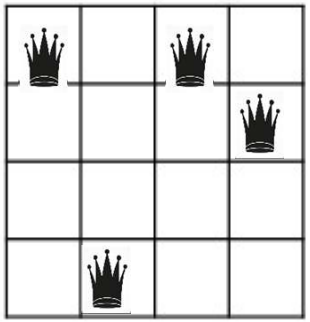
2	4	2	2	2

Probability=0.42



4	4	2	2	2

Probability=0.42



1	4	1	2	2

Probability=0.33

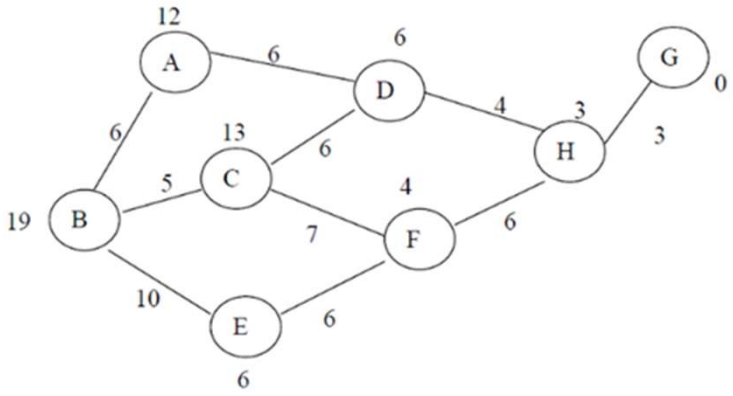
12 N = {4,2,2,3,3,2,1,3,2,1,3,2}

F(n)	Freq	Prob
1	2	2/12
2	5	5/12
3	4	4/12
4	1	1/12

We will explore more on Search techniques in the next class …

Exercise 1

- Consider the following weighted graph. The number beside each node represents its heuristic value $h(n)$,which estimates the cost from that node to the goal node G.
- The start node is B, and the goal node is G.



- Apply GBFS on this graph.
- Apply A* Algorithm on this graph.
- Compare your findings from the above two
- Check if the heuristics is admissible and consistent.

Exercise 2

- A warehouse robot has to collect three packages and reach the exit.
- The warehouse has rooms connected by corridors. Some corridors have locked doors. The robot can move from one room to an adjacent room with cost 1. To pass through a locked door, the robot must first collect the matching key.
- **Warehouse layout** Rooms:

$$S, A, B, C, D, E, F, G, X$$

where:

- S= start room
- X= exit room
- Packages are located at C, F, and G
- Key K_1 is located at B
- Key K_2 is located at E

The robot must collect all three packages at C, F, and G, then reach X.

Exercise 2 Continued

Corridor	Cost	Condition
(S-A)	1	Open
(A-B)	1	Open
(A-C)	2	Open
(B-D)	2	Requires (K1)
(C-D)	2	Open
(D-E)	1	Open
(E-F)	2	Requires (K2)
(D-G)	3	Open
(G-X)	2	Open
(F-X)	2	Open

Part A: State Representation

Define a state representation for this problem.

Your state should include:

1.Current room of the robot.

2.Packages already collected.

3.Keys already collected.

Write the initial state and one possible goal state.

Exercise 2 Continued

Part B: Relaxed Problems

A relaxed problem is obtained by removing one or more constraints from the original problem.

Consider the following relaxed versions of the warehouse problem:

- **Relaxation 1:** Ignore all locked-door constraints. The robot can pass through any corridor.
- **Relaxation 2:** Ignore package collection order. Estimate the cost only as the shortest distance from the current room to the exit.
- **Relaxation 3:** Assume the robot can collect any remaining package instantly once it enters the same connected component, but it must still move to the exit.
- **Relaxation 4:** Ignore the robot's current location and estimate only the minimum cost needed to connect all uncollected package locations and the exit.

Answer the following:

1. For each relaxed problem, explain which constraint has been removed.
2. Which relaxed problem is likely to give the weakest heuristic?
3. Which relaxed problem is likely to give the strongest heuristic?
4. Why does the solution cost of a relaxed problem give an admissible heuristic for the original problem?
5. Construct few heuristics by picking some relaxed version of the problem.

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